

OVERVIEW

8 study guide sites covering Modbus, OPC UA, DNP3, IEC 61131-3, PID Controllers, SEL RTAC, Ignition SCADA, and Wireshark. Each guide is a React/Vite SPA deployed to GitHub Pages, with quizzes, flashcards, a Code Lab, and a downloadable course PDF. The hub landing page (scada-hub.vercel.app) links all 8 guides.

FEATURE WORK

- 1. Quiz Question Hints** HIGH
Add a hint field to every quiz question in all 8 guides' quizzes.js data files, showing WHERE the answer can be found (chapter name, section heading, flashcard ID). Add a "Hint" button to Quiz.jsx that reveals the source on click. Scope: 8 guides × 600–900 questions each.
 - 2. Certification Relevance Sidebar Panel** MED
Build a new sidebar button per guide (modal or slide-out panel) containing rich certification content: cert name, exam body, link, description, and which course chapters map to which exam domains. Content removed from the syllabus PDF; this is its permanent home in the web app. Scope: 8 guides.
 - 3. DNP3 Study Guide LaTeX PDF** MED
Write, compile, and deploy the DNP3 full-length study guide PDF (equivalent to the Modbus PDF). Place at site/public/study_guide.pdf. Current sidebar PDF button returns 404. Scope: DNP3 guide only.
 - 4. Wrong-Answer Tracking + Review Missed Mode** MED
Store which question IDs a user missed per chapter (Supabase already wired). Add a "Review Missed" quiz mode that surfaces only previously incorrect questions. Scope: all 8 guides + Supabase schema.
 - 5. Spaced Repetition on Flashcards** LOW
Replace the current random shuffle with the SM-2 algorithm (~150 lines JS). Cards missed come back sooner; mastered cards surface less often. Scope: all 8 guides Flashcards.jsx.
 - 6. Timed Exam Mode** LOW
Hub-level: 60 questions, 90 minutes, randomized across all guides in a topic area (OT protocols, ICS security, etc.). Supabase for scoring and leaderboard. Scope: scada-hub + Supabase.
 - 7. Cross-Guide "See Also" Callouts** LOW
Add "See Also" callouts within chapters that link to related chapters in other guides (e.g., Wireshark Modbus chapter → Modbus guide, OPC UA Security → Wireshark OPC UA chapter). Scope: all 8 guides content pages.
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CONTENT GAPS — SYLLABUS VS. COURSEWORK

- 8. Modbus: Prerequisites + ModRSSim2 + TCP Flashcards** MED
(a) Add prerequisite statement to Home.jsx / Intro chapter. (b) Mention ModRSSim2 in the TCP or Required Software section of the web guide. (c) Expand TCP chapter flashcards (currently 3; syllabus implies 5+, covering transaction ID pipelining and Unit ID in gateways).
- 9. OPC UA: Sessions/Channels Chapter + Discovery Section** MED
(a) Add a dedicated Sessions & Channels section to Architecture.jsx: token renewal, reconnection retry, ActivateSession flow. (b) Add a Discovery section covering GetEndpoints, FindServers, GDS pattern, certificate trust flow. (c) Introduce open62541 in at least one chapter page (currently only referenced in quiz questions).
- 10. IEC 61131-3: PLCopen FB Library + Vendor Portability** MED
(a) Add a section on PLCopen standard FB signatures (TON/TOF/CTU), instance management, and library navigation. (b) Add a vendor portability section on migration barriers across Siemens/Beckhoff/SEL implementations. (c) Add a table explaining which IEC version applies to which platforms.
- 11. PID: Scilab Lab Exercises + Laplace Prerequisites** MED
(a) Add Scilab step-test simulation exercises to the Process Characterization chapter. (b) Add a note to the Digital Implementation chapter clarifying the math prerequisites (Laplace/transfer functions) so students without calculus know what to skip or review first.
- 12. DNP3: Prerequisites Statement in Web Guide** LOW
Add a brief prerequisite note to Home.jsx or the Intro chapter stating that basic TCP/IP and Modbus familiarity is assumed. All 10 topics are fully covered; this is the only gap found.
- 13. RTAC: Modbus Gateway Chapter** MED
Add a dedicated Modbus Gateway chapter (or major section) covering: DNP3-to-Modbus point mapping, register layout design, data type conversion rules, and translation failure modes. Syllabus Topic 4 ("Modbus Gateway") has no corresponding web chapter.

Also expand I/O module selection and integrate SEL-5033 virtual RTAC into the lab exercise.

- 14. **Ignition: OPC-UA Integration Chapter Depth** MED
Audit `OpCua.jsx` against syllabus Topic 3 (OPC-UA Integration). Verify PLC-to-live-tags workflow and Ignition-as-OPC-UA-server config are fully covered. Add SQL prerequisite note to `Home.jsx`. (*Audit pending.*)
 - 15. **Wireshark: SSH Remote Capture + tcpreplay** LOW
Syllabus Topic 9 (Advanced Techniques) lists SSH remote capture and tcpreplay. Verify both are in `Advanced.jsx`. Add content where missing. (*Audit pending.*)
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BUGS & INFRASTRUCTURE

- 16. **Auth: Duplicate Email Signup Error** MED
Registering with an existing email shows a generic Supabase error. Show a clear "account exists" message with a "Reset password" link. Add reset-password flow triggered from the register form error state. Scope: `scada-hub AuthPage.jsx`.
 - 17. **Quiz Reports RLS — Supabase** MED
Any logged-in user can currently read any other user's quiz history. Add Supabase row-level security (RLS) policies on the `quiz_results` table: users see only their own records; admin account has a bypass policy.
 - 18. **Mobile UX Audit** MED
Overall mobile layout is unfriendly across all 8 guides and the hub. Full audit needed: text sizes, tap targets, sidebar behavior, `LearningPath` horizontal scroll on small screens.
 - 19. **Supabase Email Confirmation Redirect** LOW
Clicking the confirmation email link sends users to a broken URL. Fix: update Site URL and Redirect URLs in Supabase Auth settings to point to `scada-hub.vercel.app`.
 - 20. **LearningPath Hover Scale Bug** LOW
Hovering a step card triggers `whileHover={scale: 1.03}` — card enlarges and text becomes hard to read. Reduce or remove the scale transform. Scope: `scada-hub/site/src/components/LearningPath.jsx`.
 - 21. **RTAC PDF Layout Overhaul** LOW
RTAC study guide PDF has bad spacing, orphaned headings, and an intro style inconsistent with the Modbus PDF template. Needs a formatting pass to align with the Modbus LaTeX template.
 - 22. **OG Image Quality Improvement** LOW
Current OG images use Python PIL with the system Helvetica font — functional but basic. Replace with a custom SVG design or Vercel's `@vercel/og` for server-side image generation at deploy time. Scope: `scada-hub public/og-image.png` + all 8 guide repos.
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UI / BRANDING

- 23. **Login Page: Company A Brand Colors** MED
The SCADA Hub login/auth page (`ScadaBackground.jsx` + `AuthPage.jsx`) uses orange as its primary accent. Adjust to a specific hex color value provided by Company A. Scope: `scada-hub/site/src/components/ScadaBackground.jsx`, `AuthPage.jsx`. Collect the target hex value before implementation.
 - 24. **Login Page: Floating Telemetry Values** MED
On the animated SCADA login background, the live telemetry tickers (PV kW, SOC%, RTT ms, MVA values) overlap with the node icons on some screen sizes. Fix: (a) move ticker boxes so they do not cover the icon, and (b) animate them with a slow float (gentle vertical oscillation) to give the background more life. Scope: `ScadaBackground.jsx` node layout and ticker positioning logic.
 - 25. **All Courses Sidebar Panel — Design Polish** LOW
The "All Courses" collapsible nav item was added to all 8 guide sidebars with guide icons and Vercel URLs. Future polish: (a) highlight the current guide in the dropdown so users know which guide they are on, (b) consider adding the guide's chapter count or question count as a subtitle under each guide name.
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COURSE AUDIT — MODBUS GAPS

- 26. **Modbus Syllabus: Add RS-485 as Standalone Topic** LOW
The syllabus lists 11 topics but the web guide and LaTeX PDF both treat RS-485 Physical Layer as a full standalone chapter (Ch 4). Add RS-485 as Topic 3 in the syllabus (between Data Model and RTU Mode), bringing the total to 12 topics aligned with the actual chapter count.
- 27. **Modbus Web: Add FC08 and FC43 to Function Codes Chapter** MED
`FunctionCodes.jsx` covers only FC01, FC02, FC03, FC04, FC05, FC06, FC15, FC16. The LaTeX PDF (Ch 8) and flashcards both include FC08 (Diagnostics) and FC43 (Encapsulated Interface Transport / MEI). Add these two codes to the web chapter so all three sources are consistent.
- 28. **Modbus Web: Add Exception Codes 10 and 11 to Exceptions Chapter** MED
`Exceptions.jsx` covers only codes 01–08. The LaTeX PDF (Ch 9) and flashcards both cover gateway codes 10 (Gateway Path

- Unavailable) and 11 (Gateway Target Device Failed to Respond). Add them to the web chapter and update the syllabus wording from “eight exception identifiers” to reflect all eleven.
29. **Modbus Web: Add Connection Management to TCP Chapter** LOW
TCP.jsx does not cover stale connections or multiple simultaneous connections. LaTeX Ch 7 covers both. Add a Connection Management section to the web chapter.
30. **Modbus Web: Add Poll Rate Overrun to Troubleshooting Chapter** LOW
The syllabus explicitly lists “poll rate overruns” as a troubleshooting topic but it does not appear in Troubleshoot.jsx. Add it as a symptom scenario (symptoms: device busy exceptions, missed polls, sluggish HMI response; cause: too many registers polled per cycle or too short a poll interval for baud rate).
31. **Modbus Web Lab: Add Gateway Exception 11 and Security Audit Scenarios** MED
LaTeX Ch 12 includes Scenario 5 (Modbus TCP Gateway Exception 11) and Scenario 6 (Security Audit Finding). Neither appears in Lab.jsx. Add both as interactive exercises or frame-decode problems to close the gap.
32. **Modbus Flashcards: Expand Security Chapter** LOW
Ch 11 Security flashcards contain only 4 cards despite the chapter covering 5 attack vectors, 5 defense-in-depth layers, and 4 NERC CIP standards. Target 8–10 cards to match coverage depth of other chapters.
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COURSE AUDIT — OPC UA GAPS

33. **OPC UA Web: Add Dedicated Discovery Chapter** MED
The syllabus lists Discovery as Topic 8 (Local Discovery Server, Global Discovery Server, FindServers, GetEndpoints, endpoint enumeration, open-discovery security risk). No Discovery.jsx page exists. The topic appears only briefly in Intro.jsx and in one flashcard (t10). Add a full web chapter covering LDS/GDS architecture, FindServers vs GetEndpoints, certificate trust during discovery, and the security risk of unauthenticated endpoint enumeration.
34. **OPC UA Flashcards: Add Troubleshooting Cards** MED
Troubleshoot.jsx is a full chapter with 7 OPC UA status codes, a 5-step diagnostic ladder, UA Expert workflow, and Wireshark analysis — but zero flashcards cover troubleshooting. Add 6–8 cards targeting the most common status codes (BadCertificateUntrusted, BadNodeIdUnknown, BadUserAccessDenied, etc.) and the diagnostic ladder sequence.
35. **OPC UA Flashcards: Add Ignition Integration Cards** LOW
IgnitionIntegration.jsx covers Ignition port 62541, the 5-step certificate trust workflow, and SEL RTAC as an OPC UA server — none of which appear in flashcards. Add 4–5 cards on Ignition-specific OPC UA behavior.
36. **OPC UA Web Security Chapter: Add CRL and Role-Based Security** LOW
Flashcards sec05 (Certificate Revocation Lists) and sec07 (role-based access: Observer, Operator, Engineer, Supervisor roles) exist but neither topic appears in Security.jsx. Align the web chapter with the flashcard content by adding subsections for CRL checking and built-in OPC UA roles.
37. **OPC UA Web Services Chapter: Add RegisterNodes** LOW
RegisterNodes/UnregisterNodes is a performance optimization service that appears in flashcard s08 and Level 3 quiz questions but is absent from Services.jsx. Add a brief section covering when and why to use RegisterNodes for high-frequency polling scenarios.
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COURSE AUDIT — DNP3 GAPS

38. **DNP3 Flashcards: Add Application Layer Chapter Cards** MED
The appLayer chapter has a full quiz question set but zero flashcards. Add 5–7 cards covering the DNP3 Application Layer: request/response fragments, function codes (READ, WRITE, SELECT, OPERATE, DIRECT OPERATE), IIN bits, sequence numbering, and fragmentation reassembly. This is the only content gap found in the DNP3 audit; all 10 syllabus topics are otherwise covered.
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COURSE AUDIT — IEC 61131-3 GAPS

39. **IEC 61131-3 Flashcards: Add Lab Chapter Cards** MED
The Lab & Practice chapter (Ch 10) has 60 quiz questions but zero dedicated flashcards. Add 6–8 cards covering the practical lab scenarios: PLC program structure, scan cycle, I/O addressing syntax, and the most common beginner mistakes caught in the quiz scenario questions.
40. **IEC 61131-3 Flashcards: Split Languages Category by Language** MED
The languages flashcard category contains ~47 cards covering four distinct chapters (Intro, Structured Text, Ladder Diagram, Function Block Diagram) — each chapter has 60 quiz questions but shares a single thin flashcard set. Split into at minimum two categories: ld-fbd (graphical languages) and st-il (text languages), and expand total card count to ~20 per language to match quiz depth.
41. **IEC 61131-3 Flashcards: Cover Scenario-Type Questions** LOW
The quiz data contains ~80 scenario questions (iec.*_sc entries, ~5 per chapter) covering applied troubleshooting and design

decisions. No flashcard category addresses scenario-style content. Add a scenarios flashcard category with 10–12 cards framing the most instructive scenario setups and their resolution logic.

COURSE AUDIT — PID CONTROLLERS GAPS

- 42. PID Flashcards: Add Cards for 5 Uncovered Chapters** **HIGH**
5 of 10 quiz chapters have zero flashcards: Closed-Loop Intro (65 Qs), Loop Fundamentals (65 Qs), P/I/D Action (65 Qs), Digital PID (65 Qs), and Troubleshooting (65 Qs). That is roughly half the quiz content with no flashcard reinforcement. Priority order: (1) P/I/D Action, (2) Troubleshooting, (3) Digital PID, (4) Intro, (5) Loop Fundamentals. Target 6–8 cards per chapter.
- 43. PID Flashcards: Align Category Names to Chapter Boundaries** **LOW**
Current flashcard categories (fundamentals, tuning, dynamics, advanced, practical) do not map 1:1 to the 10 syllabus chapters. Students cannot navigate from a quiz chapter to a matching flashcard deck. Rename or split categories so each quiz chapter has an identifiable flashcard counterpart.
- 44. PID Flashcards: Add Lab / Simulation Chapter Cards** **LOW**
Ch 10 (Simulation Lab) has 65 quiz questions but zero flashcards. Add 5–6 cards covering Scilab step-test procedure, open-loop identification, and the tuning workflow performed in the lab.
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COURSE AUDIT — SEL RTAC GAPS

- 45. RTAC Flashcards: Add Cards for 4 Uncovered Chapters** **HIGH**
4 quiz chapters have zero dedicated flashcards: RTAC Architecture/Intro (no cards), ACCELERATOR Architect Software (no cards), Tag Database (no cards), and Modbus Gateway (no cards — despite being syllabus Topic 4). Target 5–7 cards per chapter, prioritizing Gateway and TagDB since those are the heaviest operational topics.
- 46. RTAC: Add Lab Chapter Quiz Questions** **MED**
The Lab & Field Scenarios chapter (Lab.jsx) has a full web page but zero quiz questions. It is the only chapter across all 8 quiz chapters with no assessment. Add 30–40 quiz questions covering the scenario walkthroughs presented in the lab page (configuration steps, fault identification, verification checks).
- 47. RTAC Flashcards: Reconcile Hardware Chapter Scope** **LOW**
The hardware flashcard category spans content from three web chapters (Architecture, Software, TagDB) rather than mapping to a single chapter. Refactor: move hardware-specific cards to an intro category and create separate software and tagdb categories so each chapter has its own flashcard home.
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COURSE AUDIT — IGNITION SCADA GAPS

- 48. Ignition Flashcards: Add Cards for 4 Uncovered Chapters** **HIGH**
4 of 10 quiz chapters have zero flashcards: Intro/Overview (65 Qs), Ignition Gateway (65 Qs), Vision Module (65 Qs), and Alarming (65 Qs). Combined that is 260 quiz questions with no flashcard reinforcement. Priority order: (1) Gateway, (2) Alarming, (3) Vision, (4) Intro. Target 7–8 cards per chapter.
- 49. Ignition Flashcards: Reconcile Quiz/Flashcard Naming** **LOW**
Quiz chapters use scripting and alarms while flashcard categories use scripts and alarming. Students navigating from a quiz result to flashcards hit the wrong category. Standardize naming across both data files so IDs match exactly.
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COURSE AUDIT — WIRESHARK GAPS

- 50. Wireshark Flashcards: Add Advanced Techniques Chapter Cards** **MED**
Ch 9 (Advanced Techniques) has 65 quiz questions but zero flashcards. This chapter covers editcap, mergecap, tshark, dumpcap ring buffer, Lua dissector scripting, and SSH remote capture. Add 6–8 cards targeting the most testable concepts: tshark command syntax, ring-buffer capture options, the SSH remote capture one-liner, and Lua dissector registration.
- 51. Wireshark Flashcards: Add Lab Chapter Cards** **LOW**
Ch 10 (Protocol Lab) has 65 quiz questions but zero flashcards. Add 5–6 cards covering the lab workflow: filter construction for Modbus, DNP3, and OPC UA captures; common frame-decode steps; and the ICS anomaly identification checklist from the lab exercises.
- 52. Wireshark Advanced Chapter: Add tcpreplay Coverage** **LOW**
Advanced.jsx covers editcap, mergecap, tshark, and SSH remote capture but does not mention tcpreplay (PCAP replay for ICS protocol testing). The syllabus Topic 9 lists it explicitly. Add a short section on tcpreplay usage: replaying a captured PCAP against a test device, rate control, and its role in testing IDS/IPS rules against replayed OT traffic.

TRAINING FEATURE — LAUNCH CHECKLIST

- 53. Run Training Events SQL Migration** HIGH
Open Supabase Dashboard → SQL Editor and run `scada-hub/supabase/20260516_003_training_events.sql`. Creates the `training_events` table, RLS policies, unique constraint on URL, and the `course+date` index. Required before seeding or deploying the `TrainingPanel` component.
- 54. Configure Vercel Env Vars for Training Cron** HIGH
In the `scada-hub` Vercel project settings, add the following environment variables:
`CRON_SECRET` — any random string (e.g. `openssl rand -hex 32`)
`BRAVE_SEARCH_API_KEY` — free tier at `api.search.brave.com` (2000 queries/month)
`RESEND_API_KEY` — free account at `resend.com` (3000 emails/month)
`ALERT_EMAIL` — `ric3639@gmail.com` (broken URL alert destination)
`SUPABASE_SERVICE_KEY` — service role key from Supabase project settings
- 55. Enable Vercel AI Gateway on scada-hub Project** HIGH
In Vercel Dashboard → `scada-hub` project → Settings → AI Gateway: enable the gateway. This provisions the OIDC token the `api/refresh-training.js` cron uses to call `anthropic/claude-haiku-4.5` without a direct API key.
- 56. Seed Initial Training Events** HIGH
After the migration runs, seed 35 verified 2026 training events and certifications across all 8 courses:
`SUPABASE_URL=... SUPABASE_SERVICE_KEY=... node scripts/seed-training.js`
Covers: Modbus (3), OPC UA (4), DNP3 (4), IEC 61131-3 (4), PID (5), RTAC (3), Ignition (5), Wireshark (6) + certs.
- 57. Deploy All 9 Projects — Training Feature** HIGH
Redeploy all 9 projects to push the `TrainingPanel` component and Vercel cron config live. Run from each project root:
`cd site && GITHUB_PAGES= npm run build && cd .. && vercel --prod --yes`
`scada-hub` also needs: `vercel --prod --yes` from the project root (no site build needed for cron function).
- 58. Troubleshooting Lesson Plan LaTeX PDF** MED
Build a multi-chapter LaTeX PDF documenting real debugging sessions as field-engineer lesson plans. Each chapter covers one class of bug or architectural decision: symptom, what was checked, root cause, fix, and what to watch for next time. Example chapter: "Missing Icon Import Crashes React App" (Ignition Zap ReferenceError). Generate at the end of each session and accumulate chapters over time. Source lives in `scada-hub/lesson-plans/`. Distinct from the per-guide study guide PDFs — this is a *developer training document*.

PROFESSIONAL DEVELOPMENT / R&D

- 59. R&D: AI Agent Skills and Agentic Tooling** LOW
Dedicate time to developing hands-on proficiency with agentic AI tools relevant to SCADA delivery — specifically Claude AI (Claude Code, agent SDK, multi-agent orchestration), OpenAI Codex, and equivalent code-generation toolchains. Goal: identify which agentic capabilities can accelerate project delivery (code generation, automated testing, documentation, diagram synthesis) and build repeatable workflows around them. Deliverable: a short internal write-up of which tools were evaluated, what they can do in this stack, and recommended adoption candidates.